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Endocrine consequences of weight loss in obese, hyperandrogenic, anovulatory women*†

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Objective: To determine whether weight loss in obese, hyperandrogenic, anovulatory women is associated with resumption of ovulation and/or with changes in insulin, androgen, and gonadotropin concentrations.

Design: Prospective, randomized, controlled study.

Setting: University research center.

Patients: Twelve obese, hyperandrogenic, anovulatory women.

Interventions: Twelve-week weight loss program in treatment (n = 6); 12-week "waiting list" in control group (n = 6).

Main Outcome Measures: [1] Ovulation; [2] fasting insulin and glucose measurements; [3] sex hormone-binding globulin (SHBG), total and non-SHBG T concentrations; [4] LH pulse frequency, amplitude, and concentration; and [5] FSH concentration.

Results: In contrast with the control group who showed no change in weight, ovulation status, or hormone levels, women in the treatment group lost an average of 16.2 kg and showed a significant increase in SHBG, a significant decline in non-SHBG T, and a decline (though nonsignificant) in fasting insulin. Four of six subjects resumed ovulation. However, no changes were evident in LH pulse frequency or amplitude or in mean LH and FSH concentrations.

Conclusions: Weight loss in obese, hyperandrogenic, anovulatory women appears to reduce insulin and non-SHBG T concentrations despite the absence of a change in gonadotropin secretion and may lead to resumption of ovulation. Fertil Steril 1994;61:598–604

Key Words: Weight loss, obesity, polycystic ovaries, insulin, androgens, gonadotropins

The relationship between obesity and menstrual disorders has been appreciated for some time (1, 2). Moreover, weight loss in obese anovulatory women is often associated with resumption of menstrual function (3–5) and with alterations in androgen (3–5), gonadotropin (4), and insulin concentrations (4, 5). A clinical condition commonly associated with obesity and anovulation is polycystic ovarian syndrome (PCOS). Although there is intense controversy about the localization of the fundamental pathophysiologic defect(s) in PCOS (6), a consensus has emerged that PCOS is characterized by anovulation, hyperandrogenism, insulin resistance, and altered gonadotropin secretion (7, 8).

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A role for ovarian androgens in PCOS is suggested by the observation that removal or ablation of ovarian tissue often reduces circulating androgen concentrations and restores a normal, ovulatory, gonadotropin profile (9). Moreover, T is not suppressed by dexamethasone (10) but is suppressed to the castrate range by administration of a GnRH analogue (11). Insulin has been found to stimulate androgen production by ovarian stroma (12), and insulin resistance and hyperinsulinism are common among women with PCOS (13). It has been hypothesized from several lines of evidence that the direction of causation is from hyperinsulinism to hyperandrogenism, rather than the reverse (14). Finally, a consistent finding among women with PCOS is altered gonadotropin secretory profiles. Although single blood samples do not consistently show the classic elevation in LH or in the ratio of LH to FSH (15), the amplitude and frequency of LH pulses appear to be higher in women with PCOS when compared with follicular phase controls (16). It is not clear, however, whether this implies a primary hypothalamic defect in PCOS or rather a response to an altered peripheral hormonal milieu.

Given the above data and the finding that many obese, hyperandrogenic, anovulatory women who lose weight have a restoration of ovarian function (3–5), we hypothesized that weight loss would be associated with ovulation and amelioration of these endocrine abnormalities characteristic of PCOS, namely, a decline in insulin resistance, a reduction in androgens, and restoration of normal gonadotropin secretion.

MATERIALS AND METHODS

Twelve obese, hyperandrogenic, anovulatory women were recruited as subjects for the study from the clinical practice of one of the authors (D.S.G.) or from a newspaper advertisement. All subjects met the following criteria: [1] between 20 and 40 years of age; [2] absence of any medical conditions that would compromise the safety of a very low calorie diet, including chronic renal failure, cardiovascular or cerebrovascular disease, liver disease, cancer, or psychosis; [3] between 130% and 200% of ideal body weight according to the 1983 Metropolitan Height and Weight Tables for Women (17); [4] a screening serum T concentration of ≥70 ng/dL (≥2.43 nmol/L); [5] absence of specific causes of hyperandrogenism such as a tumor of the adrenal or ovary, congenital adrenal hyperplasia, Cushing's syndrome, acromegaly, hyperprolactinemia, or drug-induced hyperandrogenism; and [6] anovulation or oligo-ovulation as indicated by less than four bleeding episodes in the previous 12 months and a negative pregnancy test.

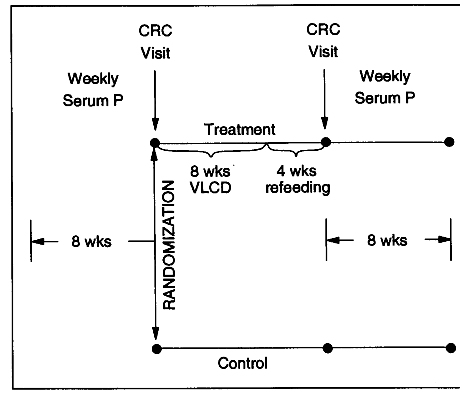

Figure 1 Schematic outline of study protocol.

In the screening phase of the study, subjects had weekly serum P levels drawn for 8 weeks. Evidence of more than one P concentration ≥ 15 nmol/L during this time interval disqualified the subject from the study; two potential subjects were not enrolled on this basis. After these serial P determinations, subjects were randomized to a treatment or control condition. The treatment condition was a 12-week behavioral weight control program, described below. The control condition was a 12-week wait-list period. All subjects were assessed before and after the 12-week period, after which they again had eight serial, weekly serum P determinations. After that time, control subjects were treated in the same behavioral weight control program. A schematic illustration of the design is shown in Figure 1.

Before initial testing, all women followed a weight-maintaining diet containing approximately 250 g carbohydrate for 3 days. They were then scheduled for an outpatient assessment session, at which height, weight, and body fat distribution (waist:hip ratio) were measured. A fasting blood sample was also obtained for measures of glucose and insulin. Subjects were then admitted as outpatients to the Clinical Research Center (CRC) of the University of Pittsburgh School of Medicine. Subjects arrived at the CRC at 8:00 A.M., an intravenous catheter was introduced into a forearm vein, and ≥30 minutes was allowed for accommodation to the catheter. Beginning at 9:00 A.M., and continuing for 12 hours, samples were obtained every 10 minutes for LH, and every 2 hours for FSH, T, sex hormone-binding globulin (SHBG), and non-SHBG T.

After the above baseline testing, all women who were randomized to the treatment condition followed a behavioral weight control program in which they were seen weekly for 12 weeks. During the initial 8 weeks of the weight-loss program, subjects were placed on a very low calorie diet, consuming 400 kcal/d (1,680 kJ/d) of lean meat, fish, or fowl, with use of a liquid formula called Optifast (Sandoz Nutrition, Minneapolis, MN) for occasional meals. This diet was supplemented with a multivitamin, calcium, and potassium. After 8 weeks on the very low calorie diet, foods were gradually reintroduced in the following order: vegetables, fruit, bread, and cereals. Calorie goals were increased by 840 kJ/wk over a 4-week period until subjects consumed 4,200 to 5,040 kJ/d.

Throughout the 12-week period, subjects were also provided with training in behavior modification to help produce long-term changes in eating habits. The behavior modification program included strategies for changing the home environment (stimulus control techniques and slowing down the act of eating), dealing with social situations, changing cognitions related to eating, preplanning, self-reinforcement, motivation, and relapse prevention. The women were also taught to increase their activity level. Gradually increasing goals for calorie expenditure were prescribed, with walking emphasized as an appropriate form of exercise. Initially, women were instructed to expend 1,050 kJ/wk in extra activity (equivalent to walking one-half mile on 5 d/wk for a 67.5-kg individual). Gradually, the goal was increased until subjects walked 2 miles/d, 5 d/wk (total of 10 miles or approximately 4,200 kJ/wk for a 67.5-kg person). Although exercise alone can affect insulin sensitivity, we chose to include exercise as a component of the weight-loss program rather than studying its independent effect. After the 12-week weight-loss program (or the 12-week waiting interval for controls), all of the assessments completed before treatment were repeated.

Serum LH concentrations were determined in duplicate by fluorescent immunoassay (FIA, Delfia, hLH Spec; Wallac, Inc., Gaithersburg, MD). The within-assay coefficient of variation (CV) is 3% to 5% across a dose range of 1 to 40 IU/L. The interassay CV is <5%. The reference standard is World Health Organization (WHO) second international standard 80/552. The sensitivity of the FIA is 0.05 IU/L. Serum FSH concentrations were determined by FIA (Delfia; Wallac, Inc.). The assay standard is the second international reference preparation 78/549. The sensitivity is 0.05 IU/L and the within-assay CV is <5%.

Total T levels were measured using a highly specific dextran-charcoal RIA after extraction of serum with diethyl ether (18). Sex hormone-binding globulin was measured by solid phase binding of tritiated T-SHBG to concanavalin A-sepharose (19). Values are expressed as nanograms of T bound per deciliter. Measurement of the non-SHBG-bound T (non-SHBG-T) fraction was based on the separation of SHBG-bound tritiated from unbound and albumin-bound T by precipitation with ammonium sulfate (20).

The pulse number per 12 h and pulse amplitude were determined by cluster analysis (21). A cluster width of 2 for nadirs and 1 for peaks was used. A t-statistic of 2.1 for both upstroke and downstroke was applied to limit the false-positive rate to 5%.

Insulin was measured by RIA (22), and glucose was measured with an autoanalyzer using the hydrogen peroxide method (Yellow Springs Instrument Company, Yellow Springs, OH).

Paired t-tests were used to compare baseline characteristics in the two groups. To analyze the effect of weight-loss on the hormonal measures, analysis of variance (ANOVA) was performed, using group (treatment or control) as one factor and time (before or after the 12-week study interval) as a second repeated-measures factor. The main hypothesis of the study was that there would be a greater change in outcome variables (i.e., insulin, androgens, and gonadotropins) in women who underwent weight loss than in wait-list controls; this was tested by examining the group $\times$ time interaction effects in these ANOVAs. All data analysis was performed using the ANOVA program for repeated measures in SAS (23). Statistical significance was defined as $P < 0.05$ under a two-tailed test.

RESULTS

Characteristics of treatment and control subjects are shown in Table 1. Women averaged 32 years of age and weighed 108.45 kg at entry. At baseline, Student's t-tests revealed no significant differences between treatment and control subjects with respect to age, percent of ideal body weight, or serum T concentration.

The 12-week behavioral weight-loss program resulted in a reduction in weight from 108.0 $\pm$ 5.3 to 91.8 $\pm$ 6.0 kg (mean $\pm$ SE), whereas the control condition was not associated with a change in weight. Repeated-measures ANOVA revealed a significant interaction effect ($F = 58.9$, $P < 0.0001$),

Table 1 Subject Characteristics*

Subject	Age	Height	Weight	Percent of Ideal Body Weight	Total T
	y	*cm*	*kg*		*ng/dL*†
Treatment (n = 6)	32.2 ± 4.9	162.5 ± 7.5	108.0 ± 13.0	175.5 ± 17.2	81.7 ± 7.8
Control (n = 6)	31.2 ± 3.9	157.5 ± 5.0	108.0 ± 13.8	181.3 ± 30.3	92.0 ± 20.4

* Values are means ± SE.

† Conversion factor to SI unit, 0.0347.

reflecting the marked reduction in weight among the treatment group in comparison with the lack of weight change among controls. Patients encountered no physical or psychological difficulties during the dieting phase.

Figure 2 shows the observed changes in total T, SHBG, and non-SHBG-T. Sex hormone-binding globulin was measured in serum pooled from the first six samples on each of the 2 study days. For total and non-SHBG-T, the mean of seven determinations over 12 hours was calculated for each subject. Sex hormone-binding globulin increased in the treatment group from 0.60 ± 0.09 μg/dL at baseline to 0.73 ± 0.09 μg/dL after weight loss. There was no corresponding change among controls before and after the wait-list period (0.50 ± 0.06 versus 0.47 ± 0.06 μg/dL). These results were reflected in a significant treatment × time interaction effect for SHBG (F = 7.81, P = 0.02). Non-sex hormone-binding globulin-T declined among weight-loss subjects (13.2 ± 2.9 versus 8.4 ± 1.6 μg/dL) but not among controls (10.4 ± 2.2 versus 11.1 ± 1.9 μg/dL), also leading to a significant interaction effect (F = 7.32, P = 0.02). The direction of differences was similar for total T, but the interaction effect did not reach statistical significance (F = 1.67, P = 0.23). Given a decline (albeit nonsignificant) in total T with an increase in SHBG, the decline in non-SHBG-T would suggest that weight loss is associated with a reduction in the production rate of T.

The effects of weight loss on glucose and insulin are shown in Figure 3. Fasting insulin was quite elevated in both groups at baseline. Fasting insulin levels declined markedly in the weight-loss treatment group (73.6 ± 7.01 versus 57.1 ± 11.5 μU/mL [conversion to SI unit, 7.175]), but changed only minimally in the control group (76.9 ± 7.86 versus 72.1 ± 10.13 μU/mL), leading to an interaction effect that approached statistical significance (F = 3.81, P = 0.08). Fasting blood glucose values were not significantly altered during weight loss, although there was a downward trend in the values among weight-loss subjects.

Although subjects in both conditions had markedly elevated waist:hip ratios, consistent with their insulin resistance (24), women in the treatment group showed a decline (0.95 ± 0.02 to 0.92 ± 0.03) whereas those in the control group showed no

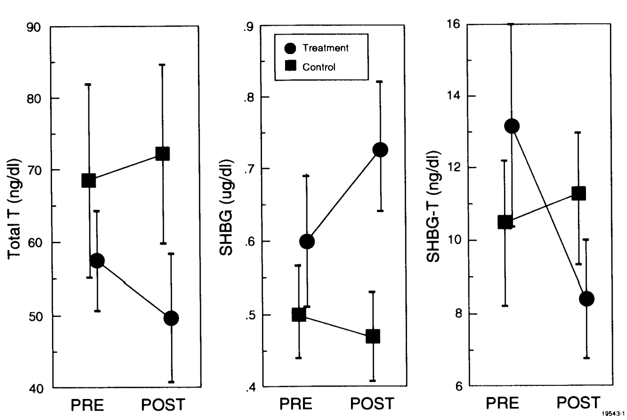

Figure 2 Total T (ng/dL), SHBG (μg/dL), and non-SHBG-T (ng/dL) for subjects in the treatment group before and after weight loss (●) and for subjects in the control group before and after the waiting list period (■). Values are means ± SE (error bars).

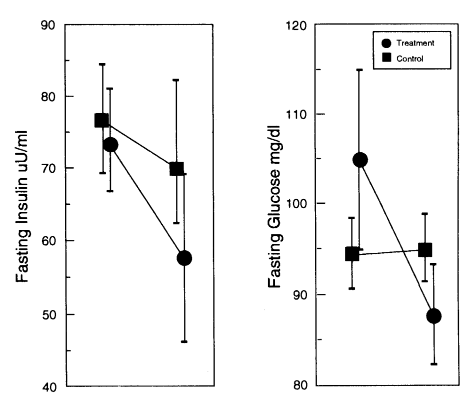

Figure 3 Fasting insulin (μU/mL) and glucose (mg/dL) for subjects in the treatment group before and after weight loss (●) and for subjects in the control group before and after the waiting list period (■). Values are means ± SE (error bars).

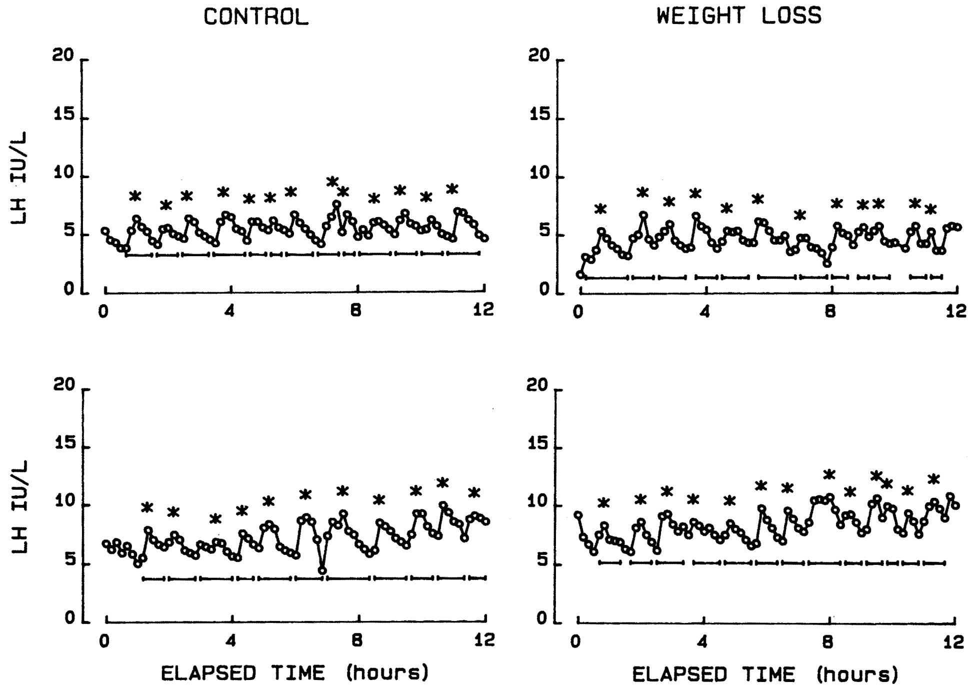

Figure 4 Representative graphs of LH secretion before (*top*) and after (*bottom*) the study interval for representative control (*left*) and weight-loss (*right*) subjects.

change (0.95 ± 0.02 to 0.95 ± 0.02). These findings, consistent with the suggestion of a decline in fasting insulin in the treatment group, were associated with a significant interaction effect ($F = 9.40$, $P < 0.05$).

Despite the change in non-SHBG-T and the suggestion of a change in fasting insulin, no changes were evident in LH pulse frequency or amplitude or in mean LH concentrations. Representative subjects from treatment and control groups are shown in Figure 4, which compares concentration profiles before treatment (*top*) with after treatment (*bottom*). Mean data for all subjects are presented in Table 2. Results for FSH, taken as mean values across the 12-hour CRC visit, were also unaffected by weight loss: mean FSH concentrations for treatment and control groups, both before and after the 12-week treatment or waiting interval, were in the range of 3.5 to 4.5 IU/L.

Detection of ovulation before and after weight loss (treatment group) or the 12-week waiting period (control group) was based on serial weekly serum P determinations for 8 weeks. Excellent compliance was achieved in obtaining weekly serum P values: all samples were obtained except for one sample each in the control and treatment groups. Although none of the subjects had more than four bleeding episodes in the previous 12 months, three subjects had one ovulatory P level during the 8-week pretreatment screening period of weekly P sampling, one in the treatment group and two in the control group. Four of the six women who underwent weight-loss treatment had ovulatory P levels after treatment, whereas only one subject in the control group had an ovulatory P after the waiting period. This was the same subject who had one ovulatory level before the waiting period.

DISCUSSION

In 1957, Mitchell and Rogers (1) reported the first systematic investigation of the relationship

Table 2 Luteinizing Hormone Results*

Group	No. of pulses per 12 hours		Pulse amplitude		LH concentration	
	Before	After	Before	After	Before	After
			mIU/mL†		*mIU/mL†*	
Treatment (n = 6)	11.5 ± 0.7	11.2 ± 1.5	2.3 ± 0.3	2.6 ± 0.4	7.4 ± 1.5	8.8 ± 2.2
Control (n = 6)	10.3 ± 1.2	9.7 ± 0.7	1.8 ± 0.2	1.9 ± 0.3	5.8 ± 0.5	6.5 ± 1.1

* Values are means ± SE.

† Conversion factor to SI unit, 1.00.

between obesity and menstrual disorders. Among 60 patients with amenorrhea, 29 (48%) were obese, defined as ≥20% above ideal body weight. In 19 additional patients with "functional uterine bleeding," which may have represented anovulatory bleeding, obesity occurred in 11 (58%). The rate of obesity in those women with ovulatory dysfunction was statistically higher than that observed in two control groups of 201 and 730 women, in which the percentage with obesity was 7.0% and 8.8%, respectively.

Although the relation between obesity and anovulation has become clinically accepted, surprisingly little corroborative data have accumulated. Green and co-workers (2) reported a case-control study involving 376 infertile women with ovulatory dysfunction and 665 fertile controls. The relative risk of infertility due to ovulatory dysfunction was 2.1 (95% confidence interval 1.0 to 4.3) among nulligravid women (n = 204) who were ≥20% above ideal weight. However, no association was observed for women (n = 172) who were previously pregnant (relative risk = 1.1). Because there was no information on the endocrine characteristics of the subjects in this study, fertile and infertile obese women could not be compared with respect to androgens, insulin, or gonadotropins.

Literature on the consequences of weight loss among obese anovulatory women is also limited. Bates and Whitworth (3) reported on 18 infertile women with PCOS who exceeded their ideal body weight by ≥20%. Thirteen of these subjects lost >15% of their body weight by dietary restriction; 10 of these women conceived. The extent of oligo-ovulation before weight loss was not documented, and no control or comparative treatment groups were examined. In 7 women for whom androgen data were available before and after weight loss, there was a significant ($P < 0.001$) reduction in both androstenedione and T, based on single blood samples. Gonadotropin and insulin concentrations were not measured in that study.

Pasquali and co-workers (4) studied the effects of weight loss in 20 obese anovulatory women, 14 of whom were thought to have PCOS. After an 8-month weight-loss program, mean body weight decreased from 86 to 76 kg. Eight of the 20 women had a significant increase in menstrual cyclicity, as inferred from menstrual histories. As in the study by Bates and Whitworth (3), there was no control group. A significant decline in T and LH was seen after weight loss, as well as a decline in insulin levels, although free T levels were unchanged. The finding of a reduction in LH concentration, which contrasts with the present findings, was based on a single blood sample drawn before and after weight loss. Moreover, the women in that study had substantially lower serum insulin levels before and after treatment than the women in our study.

Recently, Kiddy and co-workers (5) reported an analysis of 24 obese women with PCOS who underwent a low-calorie low-fat diet for 7 months. Endocrine assessments before and after the diet were compared between women who lost >5% of their prediet weight (n = 13) and those who lost ≤5% of their weight or who gained weight (n = 11). Among the group of 13 women who lost >5% of their starting weight, 9 of 11 women with amenorrhea or oligo-ovulation had an improvement in menstrual function, as measured by pregnancy or resumption of regular menstrual cycles. Moreover, this group showed a significant increase in SHBG and decrease in free T and fasting insulin levels. A subset of individuals had frequent sampling for gonadotropins. It was reported that weight loss did not influence the mean concentration, amplitude, or frequency of LH, although specific data were not provided. No changes in SHBG, T, insulin, or gonadotropins were seen in the women who did not lose weight. In the study by Kiddy et al. (5), the control group was not experimentally randomized but rather were self-selected as those who did not respond to the weight-loss program. The overall results are consistent with our present findings, however.

In the present study, we addressed some of the methodological limitations of previous investigations by incorporating a "waiting-list" control group into the design, by objective assessment of ovulation before and after weight loss, by obtaining a more complete picture of gonadotropin secretion from frequent blood sampling (every 10 minutes) over a 12-hour period and by obtaining better estimates of androgens from periodic (every 2 hours) sampling over 12 hours. An obvious limitation of this study is the small sample size. An increased sample size would allow estimates of the strength of the relationship between the magnitude of changes in endocrine parameters and the likelihood of restoration of menstrual function.

Our patients were markedly hyperinsulinemic at baseline and even after weight loss had fasting insulin levels that exceed that of obese nonandrogenic women. Prolonging the weight-control program and expansion of the sample size may be useful to determine whether the subset of women who resume menstrual function are the ones with the greatest decline in insulin and androgens. A striking finding in our study is the absence of any change in LH concentration, pulse frequency, or pulse amplitude in obese PCOS women after weight loss. Our data for the treatment group show virtually no change in these parameters before and after weight loss, quite comparable to the results in the control group.

In summary, our findings are suggestive that weight loss in women who are obese hyperandrogenic, and hyperinsulinemic is associated with a decline in non-SHBG-T and insulin concentrations and with an increased likelihood of ovulation. The absence of changes in LH secretion suggest that weight loss per se or associated endocrinologic sequelae do not directly influence the hypothalamic or pituitary regulation of LH secretion, but these data do not exclude the possibility that LH secretion or hypersecretion plays a role in the development and/or maintenance of hyperandrogenism in obese, hyperinsulinemic women. As FSH concentrations were not sampled daily, the possibility also remains that weight loss is associated with an increase in FSH secretion sufficient to support folliculogenesis (25) in those who had evidence of ovulation after treatment. Our results confirm the association between obesity, hyperandrogenism, hyperinsulinemia, and anovulation and further support the clinical notion that weight loss is an effective therapy in this subset of women.

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